

SHUTTLE MISSIONS

Whilst sorting the Target Determination tables to accompany the new Gazetteer, I knew I needed to have a way to show the shuttle missions which the Eighth Air Force flew at various times.

These rules can be used as part of a campaign, or you could just fly one to get a different experience - the crews involved with them certainly did get a wildly different time from that to which they were accustomed in the UK. They represent the shuttle mission to Regensburg in August 1943 and the various Eighth Air Force "Frantic" Missions from June to September 1944.

These missions fly to the nominated target and then on to a base in North Africa for the Regensburg mission or to one of the two bases (Poltava and Mirgorod) in Ukraine for the Frantic missions. There may then be missions your crew fly from the Ukraine base against Axis targets in Eastern Europe, otherwise there are one or two legs to come home, which usually involve bombing a target in France as part of the leg returning to the UK. I have not made extensive "what if?" tables for these intermediate missions - for simplicity I have kept them to the locations that were actually targeted.

When playing as part of a campaign, follow the actual targets for whichever Shuttle mission you are instructed to complete (Regensburg, or Frantic II, V, VI or VII). Ignore the historic dates on which these missions were carried out; these are for interest only.

If you are shot down, there are rules for the return time. Similarly, if your bomber is unrepairable whilst abroad you will be returned to the UK via a transit route.

Some of these targets are in areas where other nations' fighters will be encountered. I have included simple rules for these situations, based on the expansion module for the game. I have not (yet!) worked out a proper table 5-3 for the Axis Minor nations (Romania, Hungary, Italy) and the Luftwaffe in southern Europe, but I am collecting sources and will do it at some point.

I have tried to make this an "all inclusive" module, and hence have included the tables for you to utilise the advanced weapon model (Table 5-11, as posted separately on BGG), if you wish. At the end are the Axis aircraft multipliers if you use the original game model.

You will note that I have made Axis fighter encounters less likely when on unusual routes. This is for two reasons - firstly the Germans were not expecting these shuttle missions, and hence were tactically surprised, and secondly the deployment of their assets was to defend against the onslaught in the West; hence fewer fighters were deployed in these areas (and had huge areas to defend). This is consistent with the very scant losses suffered on these missions.

I have included a once-only German night attack. Historically this was one of the most successful German missions to destroy USAAF heavy bombers, with 47 destroyed and 17 substantially damaged. Only 9 aircraft at Poltava remained serviceable. Luckily, the attack was so well executed that there were few stray bombs on the tented accommodation area and casualties were minimal (although the Soviet fire and damage control crews were forced to work during the raid, and did suffer losses). One could argue that Soviet advances put the Ukraine bases beyond the reach of German aircraft after mid-July, but I have retained the raid as a possibility for every mission (although limiting it to a once-only event).

I am amazed how much material this took to get organised, but I have tried to make it a stand-alone as possible, so can easily be referenced when needed, and set aside for most of the game.

Please also see the attached document if you want the background data I extracted for all of the Frantic missions. As always, some sources conflict (even internally), but you can see all of the Frantic missions flown by both the Eighth and 15 Air Forces in the summer of 1944.

Chris Riches

OPERATION FRANTIC - SUMMARY OF MISSIONS

Operation	AF	Date	From	To	Target
Frantic I	15th	2 Jun 44	Italy	USSR	Debrecen M/Y, Hungary
		6 Jun 44	USSR	USSR	Galati AF, Romania
		11 Jun 44	USSR	Italy	Focsani AF, Romania
Frantic II	8th	21 Jun 44	UK	USSR	Ruhland oil refinery, Germany
		26 Jun 44	USSR	Italy	Drohobycz (refinery and adjacent M/Y), Poland
		3 Jul 44	Italy	Italy	Arad M/Y, Romania
		5 Jul 44	Italy	UK	Beziers M/Y
Frantic III	15th	22 Jul 44	Italy	USSR	Buzau AF, Romania Zilistea AF, Romania
		25 Jul 44	USSR	USSR	Mielec AF, Poland
		26 Jul 44 (part 1)	USSR	Italy	Airfields in the Bucharest-Ploesti area
		29 Jul 44 (part 2)	USSR	Italy	Keckskemet AF, Hungary
Frantic IV	15th	4 Aug 44	Italy	USSR	Focsani AF, Romania
		6 Aug 44	USSR	Italy	Craiova M/Y and other railroad targets in the Bucharest-Ploesti area
Frantic V	8th	6 Aug 44	UK	USSR	Rahmel A/C Factory
		7 Aug 44	USSR	USSR	Trzebinia oil refinery, Poland
		8 Aug 44	USSR	Italy	Buzau AF, Romania Zilistea AF, Romania
		12 Aug 44	Italy	UK	Toulouse Francazal AF
Frantic VI	8th	11 Sep 44	UK	USSR	Chemnitz Industrial Area
		13 Sep 44	USSR	Italy	Diosgyor, Hungary - steel and armament works
		17 Sep 44	Italy	UK	None
Frantic VII	8th	18 Sep 44	UK	USSR	Warsaw supply drop
		19 Sep 44	USSR	Italy	Szolnok, Hungary M/Y
		23 Sep 44	Italy	UK	None
Frantic VIII	8th	30 Sep 44 2 Oct 44 3 Oct 44	UK	USSR	Warsaw supply drop. All postponed, then cancelled.

GAME NOTES

Targets

When playing as part of a campaign, follow the actual targets for whichever Shuttle mission you are instructed to complete (Regensburg, or Frantic II, V, VI or VII). Ignore the historic dates on which these missions were carried out; these are for interest only.

Zone Tables

For a shuttle mission, the zone in which the target is located is indicated by dark green shading.

Target	Zone 15
Bordeaux	-1 / F

Friendly territory is indicated by light green shading. This is relevant for any crash landing and bail-out attempts.

Target	Zone 30
Ruhland	-3 / U

There is one extra country code: Alg = Algeria

Special Rules

If the 5-3 tables have different nationalities listed, then use all German fighters and then either the Italian, Hungarian or Romanian according to which country's territory the Zone is in.

23	None	None; Disrupted Formation	Fw 190: VD (b) It: MC.205: 6Lo Hu: Me 210: 6Lo Ro: Bf 109: 6Lo
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e.g.: On Heavy, the attackers would be one German Fw 190 and then one of the other Axis nation's aircraft, depending on whose territory you were flying over.

If under 10,000' an aircraft will always be subject to a Light strength flak attack in any Ukraine zone (even if friendly). This is due to rigid flight corridors / timings and a "shoot first" instruction to anti-aircraft units.

On the last leg back to the UK, the Zone tables have an extra row labelled Escort (Fair, or Good). Roll a D10 and if the result is within the stated range then that level of fighter cover occurs. If you fail the roll, you get one level of cover worse (Fair becomes Poor; Good becomes Fair).

Target	Zone 16	Zone 17	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Bordeaux	-1 / F	-1 / F	-1 / F	-1 / F	-1 / W-F	-1 / W	-2 / W
Escort (Fair) d10:	-	-	-	-	1-5	1-6	1-6

e.g.: From Bordeaux, you would roll for fighter cover in Zone 20, and get Fair cover on a roll of 1-5.

Target	Zone 23	Zone 24	Zone 25	Zone 26	Zone 27		
Beziers	-2 / W	-2 / W	NA / E	UK Base			
Escort (Good) d10:	1-5	1-5	1-5				

e.g.: From Beziers, if you rolled a 6 in Zone 24, you would get Fair cover. If you had rolled a 5, you would have got Good cover.

References

Reluctant Allies: The United States Army Air Force and the Soviet Voenno Vozdushnie Sily; Daniel P. Bolger; University of Chicago

[Reluctant Allies: The United States Army Air Force and the Soviet Voenno Vozdushnie Sily 1941-1945, \(dtic.mil\)](#)

This is a good source of statistical information. It also includes the routes flown by the various Frantic missions.

Fighting with the Soviets: The Failure of Operation Frantic; Mark J. Conversino

This has more background, and is really centred on the politics and diplomacy. It adds to the Bolger work from a depth and rounding out the bigger picture, but is quite light on statistical data. That said, it is an absorbing read and presents a good narrative of the whole Frantic story.

Tactical Mission Reports

These are the USAAF reports on operations. Two of them are readily available online in decent quality: 21st Jun to 5th July

[Eighth Air Force tactical mission report : operation no. report of shuttle operation, 21 June - 5 July. - World War II Operational Documents - Ike Skelton Combined Arms Research Library \(CARL\) Digital Library \(oclc.org\)](#)

6th to 12th Aug

<https://cgsc.contentdm.oclc.org/digital/collection/p4013coll8/id/956/rec/3>

The Poltava Affair, Glen B. Infield

This is the first book on the topic and has nothing to recommend it vs the Conversino book. It deals with the politics and diplomacy in a more superficial fashion, leaving you underwhelmed and feeling the author wanted to rush it into publication rather than do a thorough job.

Operation Frantic: The Shuttle Missions to Russia; James A. Oliveri.

Avoid this poor quality book. It lacks coherence, oversight of the big picture, detail on the missions, and accuracy. It is a jumble of reminiscences with a vague thread connecting them to Frantic.

STANDARD PROCESS WHEN ARRIVING AT A SHUTTLE BASE

1	Crew Status	If the crew bailed out, they are repatriated back to the UK. See step 5. Otherwise, roll for Wounds and recovery time as normal.
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WOUND RECOVERY			
d6	1 Light Wound	2 Light Wounds	Serious Wound
1	No effect	No effect	Miss 1d6 days
2	No effect	No effect	Miss 2d6 days
3	No effect	No effect	Miss 3d6 days
4	No effect	Miss 1d6 days	Stable; roll again in 7 days *
5	Miss 1d6 days	Miss 2d6 days	Recovers but fly no more *
≥ 6	Miss 2d6 days	Incapacitated. Roll after 10 days on SW column	Fatal

- +1 B-24 wrecked in landing on land (Table 7-1) for all forward crew (Nose, Pilot Compartment, Radio Room)
- +1 Bailed out, or crash-landed in Ukraine away from the intended landing location (Medical facilities were good at Mirgorod and Poltava, but more rudimentary elsewhere)

* These results automatically result in the crewman being repatriated back to the UK (see step 6).

2	Time until next mission	Roll on the table below. All crew which are fit on the date of the next mission are eligible to take part. Others are left behind and are repatriated after then end of their convalescence period (see step 6). If there is an "out and back" mission from a remote base (Frantic V from Ukraine and Frantic II from Italy) then the crew member could miss the intervening mission and still be part of the crew for the onward journey.
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D10	Days
1-2	1
3-4	2
5	3
6	4
7	5
8	6
9	7
10	8
11	9
12	11
13	13

- +1 Outbound mission from the UK (all Frantic missions)
- +3 Regensburg outbound mission (Aircraft landed at a variety of unprepared locations and had to be corralled back into a coherent force to fly back to the UK)

3	German Night Raid	If the mission has just landed at a base in Ukraine, roll a D10: 1-8 No event 9-10 German Night Attack If the German Night Attack occurs, it is a once-only event. Do not roll again on future missions. Make another D10 roll: 1-6 Aircraft is destroyed. Repatriate back to the UK - see Step 5. 7-9 Aircraft is repairable in D10 days. Go to Step 4. 10 Aircraft is undamaged. Go to Step 4.
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4	Determine Repair Time	Conditions in North Africa and Ukraine were much more difficult than in the UK or Italy, hence use the following table for repair times (based on using Peckham Points) when in North Africa or Ukraine. Add to this any damage suffered from the night raid.
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AIRCRAFT WRITEOFF	
Points Damage	Repair %
0-199	Always repairable
200-209	01-50
210-219	01-42
220-229	01-31
230-239	01-19
240-249	01-11
250+	Nil - Cat-E writeoff

REPAIR TIMES	
Points Damage	Time
01-49	Overnight
50-99	2 days
100-149	3 days
150-199	4 days
200-249	5 days

+1 day Two or more engines out
+1 day Both wing roots have three or more hits
+1 day For each tail flying surface that has three hits
+1 day For every +2 landing modifier

5	Aircraft Condition	If the aircraft is unrepairable All the fit crew may be assigned to other aircraft to make up for wounded members of their crew. Roll a D10 for each fit crew member: 1-5 Fly back to the UK on a transport flight (see Step 6). 6-10 Assigned to a position on another returning aircraft (the crewmen are all on different aircraft). Fly the return mission with your senior person. The other crew arrive safely.
		If aircraft is repairable, but not in time for the mission Return as a non-combat mission. Determine when aircraft is repaired; fly on this date + D10/2 (round down) days [to allow for good weather]. Do not fly the mission; roll % dice: on a 99-100 you hit a mechanical problem, return to the take-off base and try again in D6/2 (round down) days, with another mechanical failure roll. Once home, return to normal duties. If there is an "out and back" mission from a remote base (Frantic V from Ukraine and Frantic II from Italy) then the crew miss the intervening mission and if the aircraft is repaired in time will take part in the next onward journey.
		If the aircraft is repairable in time for the mission Fly the mission as normal. If any crew are unable to join the mission due to wounds, assume their position is manned by survivors from other aircraft; assign a temporary crewman to each vacant position. The unfit crew member will be Repatriated (see Step 6).
6	Repatriation	Roll for each group of crewmen that become eligible to repatriate on the same day (so if you crash-landed and crew are not utilised on other aircraft, they roll as one group).

REPATRIATION TABLE	
D10	Days
1-2	4
3-4	6
5-6	9
7-8	12
9-10	15
11	20
12	25

+2 From Russia

REGENSBURG

MISSION OUTWARD FROM THE UK		
Operation	Historic Date	Target
Return	17 Aug 43	Regensburg Messerschmitt aircraft factory

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Regensburg	-2 / W	-1 / W-N	-1 / B	-1 / B-N	0 / G	0 / G	0 / G

Target	Zone 9	Zone 10	Zone 11	Zone 12	Zone 13	Zone 14	Zone 15
Regensburg	0 / G	0 / G	+1 / G	-1 / G	-1 / G	-1 / G-Au	-2 / Au-I

Target	Zone 16	Zone 17	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Regensburg	-2 / I	-1 / I	-1 / I	-1 / I-W	-2 / W	-2 / W-C	-2 / W-I

Target	Zone 23	Zone 24	Zone 25	Zone 26	Zone 27	Zone 28	Zone 29
Regensburg	-2 / W	-3 / W	-3 / W	-3 / W	-3 / W	-3 / W	NA / Alg Base

SPECIALS	
Enemy Fighters	When in Zones 2-15, use the normal 5-3 tables. When in zones 16-28, use the Regensburg Southern Europe 5-3 tables for Axis fighter generation.
Escort strength	Use the normal Zone coverage tables for as far as it can get you; there is no fighter cover after this.
Zone comment	I have put C (Corsica) and I (Sardinia) in Zones 21 and 22 in order to give a potential bail-out location over land.

MISSION BACK TO THE UK		
Operation	Historic Date	Target
Return	24 Aug 43	Bordeaux Merignac AF

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Bordeaux	NA / W	-3 / W	-3 / W	-3 / W	-3 / W	-3 / W	-3 / W

Target	Zone 9	Zone 10	Zone 11	Zone 12	Zone 13	Zone 14	Zone 15
Bordeaux	-3 / W	-3 / W	-2 / W-F	-2 / F	-2 / F	-2 / F	-1 / F

Target	Zone 16	Zone 17	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Bordeaux	-2 / F	-2 / F	-2 / F	-1 / F	-1 / W-F	-1 / W	-2 / W
Escort (Fair) d10:	-	-	-	-	1-5	1-6	1-6

Target	Zone 23	Zone 24	Zone 25				
Bordeaux	NA / E	NA / E	UK Base				

SPECIALS	
Enemy Fighters	When in zones 2-15, use the Regensburg Southern Europe 5-3 tables for Axis fighter generation. Use the normal tables thereafter.
Escort strength	No initial escort; roll for UK escort for Zones 20+.

OPERATION "FRANTIC"

Use the normal tables for the route out to each of these targets. Listed below are the flight zones from the targets to the bases in Ukraine (USSR). Again, the zone modifiers after the target are lower, to reflect the paucity of defending fighters capable and organised to attack Eighth Air Force formations.

1. OUTBOUND MISSION FROM THE UK TO UKRAINE		
Operation	Historic Date	Target
Frantic II	21 Jun 44	Ruhland oil refinery
Frantic V	6 Aug 44	Rahmel A/C Factory
Frantic VI	11 Sep 44	Chemnitz Industrial Area
Frantic VII	18 Sep 44	Warsaw supply drop

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Chemnitz	-2 / W	-1 / W	-1 / W-N	-1 / N	-1 / N-G	0 / G	0 / G
Rahmel	-2 / W	-1 / W	-1 / W	-1 / W	-1 / W	-1 / W	0 / W-G
Ruhland	-2 / W	-1 / W	-1 / W-N	-1 / N	-1 / N-G	0 / G	0 / G
Warsaw	-2 / W	-1 / W	-1 / W-N	-1 / N	-1 / N-G	0 / G	0 / G

Target	Zone 9	Zone 10	Zone 11	Zone 12	Zone 13	Zone 14	Zone 15
Chemnitz	0 / G	0 / G	0 / G	-1 / G	-1 / Cz	-1 Cz-G	-1 / G
Rahmel	0 / G	+1 / W-G	0 / W-G	0 / W	0 / W	0 / W-G	0 / P-W
Ruhland	0 / G	0 / G	0 / G	0 / G	-1 / G	-2 / G	-2 / G
Warsaw	0 / G	0 / G	+1 / G	0 / G	0 / G	-1 / G	-1 / P

Target	Zone 16	Zone 17	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Chemnitz	-2 / G	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P
Rahmel	-1 / W-P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P
Ruhland	-2 / G-P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P
Warsaw	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P

Target	Zone 23	Zone 24	Zone 25	Zone 26	Zone 27	Zone 28	Zone 29
Chemnitz	-2 / P	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U
Rahmel	-3 / P-U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U
Ruhland	-2 / P	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U
Warsaw	-2 / P	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U

Target	Zone 30	Zone 31	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Chemnitz	NA / U Base						
Rahmel	-3 / U	NA / U Base					
Ruhland	-3 / U	NA / U Base					
Warsaw	NA / U Base						

SPECIALS	
Enemy Fighters	Use the normal 5-3 tables for Axis fighters.
Escort strength	Use the normal Zone coverage tables out to Zone 13 or the target Zone, whichever is first. After that, Fighter Cover is automatically Fair.

2. MISSION FLOWN FROM AND RETURNING TO UKRAINE		
Operation	Historic Date	Target
Frantic II	-	None
Frantic V	7 Aug 44	Attack oil refinery at Trzebinia, Poland - Kontinentale Oel (Malopolska)
Frantic VI	-	None
Frantic VII	-	None

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Trzebinia	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U-P

Target	Zone 9	Zone 10	Zone 11	Zone 12	Zone 13	Zone 14	
Trzebinia	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	-2 / P	

SPECIALS	
Enemy Fighters	Use the normal 5-3 tables for Axis fighters.
Escort strength	Fighter cover is automatically Fair for the whole mission.
Mission	This is a normal (i.e. "out and back") mission flown from the USSR and returning there.

3. MISSION FROM UKRAINE TO ITALY		
Operation	Historic Date	Target
Frantic II	26 Jun 44	Drohobycz (refinery and M/Y adjacent), Poland
Frantic V	8 Aug 44	38 A/C sent to Buzau AF, Romania 35 A/C sent to Zilistea AF, Romania
Frantic VI	13 Sep 44	Diosgyor, Hungary - steel and armament works
Frantic VII	19 Sep 44	Szolnok, Hungary M/Y

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Buzau	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U-R	-2 / R	-2 / R
Diosgyor	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U-P
Drohobycz	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U-P
Szolnok	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U
Zilistea	-3 / U	-3 / U	-3 / U	-3 / U	-3 / U-R	-2 / R	-2 / R

Target	Zone 9	Zone 10	Zone 11	Zone 12	Zone 13	Zone 14	Zone 15
Buzau	-2 / R	-2 / R	-1 / R	-1 / R	-1 / R	-1 / R-B	-1 / Y
Diosgyor	-2 / P	-2 / P	-2 / P	-2 / H	-2 / H	-1 / H	-1 / H
Drohobycz	-2 / P	-2 / P	-1 / P	-1 / P-H	-1 / H	-1 / H	-1 / H
Szolnok	-3 / U	-3 / U-R	-2 / R	-2 / R-H	-1 / H	-1 / H	-1 / H
Zilistea	-2 / R	-2 / R	-1 / R	-1 / R	-1 / R	-1 / R-Y	-1 / Y

Target	Zone 16	Zone 17	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Buzau	-1 / Y	-1 / Y	-1 / Y	-2 / Y	-2 / W	-2 / W	NA / I Base
Diosgyor	-1 / H	-1 / H-Y	-1 / Y	-1 / Y	-1 / Y	-2 / W-Y	-2 / W
Drohobycz	-1 / Y	-1 / Y	-1 / Y	-1 / Y	-1 / Y	-2 / W	-2 / W
Szolnok	-1 / H	-1 / H-Y	-1 / Y	-1 / Y	-1 / Y	-2 / W-Y	-2 / W
Zilistea	-1 / Y	-1 / Y	-1 / Y	-2 / Y	-2 / W	-2 / W	NA / I Base

Target	Zone 23						
Buzau							
Diosgyor	NA / I Base						
Drohobycz	NA / I Base						
Szolnok	NA / I Base						
Zilistea							

SPECIALS	
Enemy Fighters	Use the normal 5-3 tables in friendly (Ukrainian/Polish) zones, and the Summer 44 Southern Europe 5-3 tables for all non-friendly or water Zones.
Escort strength	Roll for Fighter Cover in every Zone either fully or partly in Yugoslavia: 1-2 Good 3-10 Fair Over Water at the end of the mission all Fighter Cover is Good. For all other Zones, cover is automatically Fair.

4. MISSION WITH 15TH AIR FORCE FROM ITALY		
Operation	Historic Date	Target
Frantic II	3 Jul 44	Arad M/Y, Romania
Frantic V	-	None
Frantic VI	-	None
Frantic VII	-	None

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Arad	-2 / W	-1 / W-Y	0 / Y	0 / Y	0 / Y	0 / Y	0 / Y

Target	Zone 9	Zone 10					
Arad	0 / R	0 / R					

SPECIALS	
Mission	This mission is carried out as a normal 15th Air Force mission. I suggest using the Summer 44 Southern Europe 5-3 tables for Axis fighter generation.

5. MISSION BACK TO THE UK		
Operation	Historic Date	Target
Frantic II	5 Jul 44	Beziers M/Y
Frantic V	12 Aug 44	Toulouse Franczal AF
Frantic VI	17 Sep 44	None
Frantic VII	23 Sep 44	None

Target	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6	Zone 7	Zone 8
Beziers	-2 / I	-2 / W	-2 / W	-2 / W	-2 / W	-2 / W-C	-2 / W
Toulouse	-2 / I	-2 / W	-2 / W	-2 / W	-2 / W	-2 / W-C	-2 / W

Target	Zone 9	Zone 10	Zone 11	Zone 12	Zone 13	Zone 14	Zone 15
Beziers	-2 / W	-2 / W	-2 / W	-2 / F	-1 / W-F	-2 / F	-2 / F
Toulouse	-2 / W	-2 / W	-2 / W	-2 / F	-1 / W-F	-1 / F	-1 / F

Target	Zone 16	Zone 17	Zone 18	Zone 19	Zone 20	Zone 21	Zone 22
Beziers	-2 / F	-2 / F	-2 / F	-2 / F	-1 / F	-1 / F	-1 / F-W
Escort (Good) d10:	1-2	1-2	1-3	1-3	1-3	1-4	1-5
Toulouse	-2 / F	-2 / F	-2 / F	-2 / F	-2 / F	-1 / F	-1 / F
Escort (Good) d10:	-	1-2	1-2	1-3	1-3	1-3	1-4

Target	Zone 23	Zone 24	Zone 25	Zone 26	Zone 27		
Beziers	-2 / W	-2 / W	NA / E	UK Base			
Escort (Good) d10:	1-5	1-5	1-5				
Toulouse	-1 / F-W	-2 / W	-2 / W	NA / E	UK Base		
Escort (Good) d10:	1-5	1-5	1-5	1-5			

SPECIALS	
Enemy fighters	When in zones 2-13, use the Summer 44 Southern Europe 5-3 tables for Axis fighter generation. Use the normal tables thereafter.
Escort strength	Use the 15th Air Force escort for the start of the mission until Zone 13, and the Escort rows above for the final escort into the UK. In all other Zones the escort is automatically Fair.

AXIS FIGHTERS - REGENSBURG MISSION - SOUTHERN EUROPE

5-3SA AXIS FIGHTER APPEARANCE TABLE: REGENSBURG SOUTHERN EUROPE			
#	Light	Moderate	Heavy
11	None	None	None
12	None	None	None
13	None	None	None; Disrupted Formation
14	None	None	Bf 109 (h): 6H
15	None	None	Fw 190: VD (d)
16	None	None	Bf 110: VC (e)
21	None	None	Bf 109 (h): 6H
22	None	None	Fw 190 (h): 10:30H
23	None	None; Disrupted Formation	Bf 109: VD (d)
24	None	Fw 190: 12H	Two Bf 109: 12Lv; 1:30H
25	None	Bf 110 (h): 6Lo	Two Bf 110: 3Lv; 10:30Lo
26	None	Ju 88C: 3Lv	Two Fw 190 (h): 10:30H; 3Lv
31	None (c)	Fw 190 (h): 6Lo	Two Bf 110: 6Lv; 9Lo
32	None (c)	Fw 190 (h): 6H	Me 109 (h): 3Lo Bf 110: 1:30Lo
33	None (c)	Fw 190 (h): 9Lo	Two Bf 109: 1:30H; 9Lv
34	None; Disrupted Formation	Fw 190: 6Lo	Three Fw 190: 12H; 3Lv; 3H
35	Fw 190: 1:30H	Two Fw 190: 12H; 10:30H	Three Bf 109: 12Lv; 12H; 1:30Lv
36	Bf 110: 6Lo	Two Me 210: 6Lo; 1:30Lo	Three Fw 190: 12H; 1:30H; 3H
41	Me 410: 12H	Two Fw 190: 10:30Lo; 1:30H	Fw 190: 12H Two Bf 109: 1:30Lv; 3Lv
42	Bf 110: VC (e)	Two Bf 109: 1:30H; 9Lv	Three Bf 110: 12Lv; 10:30Lv; 6Lo
43	Re.2001: 1:30H (It)	Two MC.205: 1:30H; 9Lv (It)	Fw 190: 10:30H Bf 110: VC (e) MC.205: 1:30Lv (It)
44	Bf 109: 12H	Two Fw 190: 12H; 1:30H	Three Fw 190: 10:30H; 12H; 1:30H
45	Bf 109: 1:30H	Two Bf 109: 9Lv; 12H	Three Fw 190: 3H; 9H; 12H
46	MC.202: 6H (It)	Bf 109: 1:30Lo Re.2002: 3H (It)	Two Bf 109: 1:30H; 9Lv MC.202: 12Lv (It)
51	MC.200: 12H (It)	Two Me 210: 1:30Lo; 1:30Lv MC.202: VC (e) (It)	Two Bf 109: 10:30Lv; 12Lv Two Me 210: 10:30Lv; 10:30Lo
52	Two Fw 190: 1:30H; 3H	Three Fw 190: 3H; 9H; 12H	Two Fw 190: 12Lv; 1:30Lo Two Me 410: 6Lv; 6H
53	Two Bf 109: 6Lv; 1:30H G50: 9Lv (It)	Two Bf 110: 12Lv; 6Lo Re.2001: VC (e) (It)	Four Bf 109: 12H; 12Lv; 10:30H; 1:30H
54	Two Bf 109: 12Lv; 12H	Two Bf 109: 6H; 12Lv Ju 88C: 9Lo	Two Bf 109: 12H; 3H Two Bf 110: 9H; 9Lv
55	Two MC.202: 10:30Lv; 9Lv (It)	Three Fw 190: 12Lv; 1:30H; 12H	Me 210: 6Lv Three Bf 109: 12Lv; 12H; 10:30Lv
56	Ju 88C: 10:30H MC.205: 1:30H (It)	Two Bf 109: 3Lv; 9Lv Two Bf 109: 12H; VD (b) (It)	Four Fw 190: 12Lv; 1:30Lv; 3Lo; 9H

5-3SA AXIS FIGHTER APPEARANCE TABLE: REGENSBURG SOUTHERN EUROPE

#	Light	Moderate	Heavy
61	Fw 190: 12Lo MC.202: 6H (It)	Me 210: 10:30L Me 410: 12H Two MC.202: 6Lo; 6H (It)	Three Bf 109: 12Lv; 3H; 1:30H Two MC.202: 6H; VD (b) (It)
62	Two Bf 110: 3Lo; 6Lo Me 210: 10:30H	Fw 190: 12H Bf 109: 12Lo Two Me 210: 6Lo; 9H	Four Bf 109: 12Lv; 12Lo; 12H; 10:30Lv Re.2005: 3H (It)
63	Three Bf 109: 10:30H; 10:30Lv; 10:30Lo	Four Bf 109: 12H; 12Lv; 12Lo; VD (d)	Two Fw 190: 12H; 1:30Lo Two Me 210: 3H; VC (e) Bf 109: 6H (It)
64	Three Fw 190: 6H; 3H; 12Lo	Three Bf 110: 1:30Lo; 9Lv; VC (e) Me 410: 6Lv	Five Fw 190: 12H; 10:30Lv; 9Lv; 6H; VD (b)
65	Bf 109: 12Lv MC.205: 9Lv (It) Bf 109: 1:30H (It)	Three Bf 109: 3Lo; 9H; 12H Me 410: 6H	Four Fw 190: 12H; 1:30H; 3H; 6H G.55: 6H (It)
66	Random Event		

- b Bomber cannot fire on this fighter, nor can the escort remove it.
c Fighters are driven off by other bombers. Re-roll if out of formation.
d Top Turret and Radio Room MG may fire. Escort cannot remove this attacker.
e Only the Ball Turret may fire. Escort can remove this attacker.
h If over Germany, substitute Ju 88C.

5-7 EFFECTS ON ATTACKING FIGHTERS - OTHER AXIS AIRCRAFT

[illegible]

AXIS FIGHTERS - SHUTTLE MISSIONS (6-9/44) - SOUTHERN EUROPE

5-3SB AXIS FIGHTER APPEARANCE TABLE: SUMMER 44 SOUTHERN EUROPE			
#	Light	Moderate	Heavy
11	None	None	None
12	None	None	None
13	None	None	None; Disrupted Formation
14	None	None	Fw 190 (h): 10:30H
15	None	None	Bf 109: VD (d) It: G.55: 9Lo Hu/Ro: Bf 109: 9Lo
16	None	None (f)	Bf 109 (h): 3Lv Bf 110: 1:30Lv
21	None	None (g)	Bf 109 (f) (h): 6H It/Hu/Ro: Bf 109: 9H
22	None	None (g)	Two Bf 109: 1;30H; 9Lv
23	None	None; Disrupted Formation	Fw 190: VD (b) It: MC.205: 6Lo Hu: Me 210: 6Lo Ro: IAR 81: 6Lo (r)
24	None	Fw 190(h): 12Lv	Two Bf 110 (h): 9Lv; 10:30Lo
25	None	Fw 190: 6Lv	Bf 110: VC (e) It: Bf 109: 10:30Lo Hu: Me 210: 10:30Lo Ro: IAR 81: 10:30Lo (r)
26	None	Two Ju 88C: 1:30Lo; 3Lv	Two Fw 190 (h): 10:30H; 3Lv
31	None	Two Me 410: 12H; 1:30H	Two Fw 190: 12Lv; 1:30Lo
32	None	Fw 190: 12H It/Hu/Ro: Bf 109: 1:30Lv	Bf 109: 12H It/Hu/Ro: Bf 109: 6Lo
33	None (c)	Two Fw 190: 10:30Lo; 1:30H	Two Bf 109: 12Lv; 1:30H
34	None (g)	Fw 190: 3Lo It: Re.2005: 3H Hu: Bf 109: 3H Ro: IAR 81: 3H (r)	Fw 190: 9H It: MC.205: 3Lv Hu: Bf 109: 1:30Lv Ro: IAR 81: 1:30Lv (r)
35	None; Disrupted Formation	Two Bf 109: 6Lv; 6H	Three Fw 190: 12H; 1:30H; 3H
36	Bf 110: 1:30Lv	Fw 190: 9H It: MC.205: 3Lv Hu: Me 210: 1:30Lv Ro: IAR 81: 1:30Lv (r)	Two Bf 110: 6Lv; 9Lo It/Hu/Ro: Bf 109: 12H
41	Bf 109: 12Lv	Two Bf 110: 3Lo; 6Lo	Three Fw 190: 12H; 3Lv; 3H
42	Bf 109: 10:30H	Fw 190: 12H It/Hu/Ro: Bf 109: 9Lv	Three Bf 109: 12Lv; 12H; 1:30Lv
43	Fw 190 (h): 12Lv	Two Fw 190: 10:30H; VD (d)	Three Bf 109: 1:30Lv; 9H; 12H
44	Fw 190: 6Lv	Two Bf 109: 1:30H; 9Lv	Fw 190: 12H Two Bf 109: 1:30Lv; 3Lv
45	Fw 190: 12H It/Hu/Ro: Bf 109: 1:30Lv	Two Bf 110 (h): 9Lv; 10:30Lo	Two Bf 109: 10:30Lv; 12Lv Bf 110: 10:30Lo
46	Two Bf 110: 6Lo; VC (e)	Three Bf 109: 1:30Lv; 9H; 12H	Two Fw 190: 10:30H; 1:30Lo Me 410: VC (e)
51	Two Fw 190: 12H; 12Lv	Two Fw 190: 12Lv; 3H Me 410: 12H	Three Me 110: 12Lv; 10:30Lv; 6Lo
52	Two Fw 190: 12Lv; VD (d)	Fw 190: 12H Two Bf 109: 6H; VD (d)	Three Fw 190: 10:30H; 12H; 1:30H

5-3SB AXIS FIGHTER APPEARANCE TABLE: SUMMER 44 SOUTHERN EUROPE			
#	Light	Moderate	Heavy
53	Two Bf 109: 12H; 10:30H	Three Fw 190: 3H; 9H; 12H	Four Bf 109: 12H; 12Lv; 10:30H; 1:30H
54	Ju 88C: 6Lo It: G.55: 6H Hu: Bf 109: 6H Ro: IAR 81: 6H (r)	Two Bf 109: 9Lv; VC (e) Me 210: 6H	Four Bf 109: 12H; 3H; 9H; 9Lv
55	Fw 190: VD (d) It: MC.205: 12H Hu: Me 210: 12H Ro: Bf 109: 12H	Three Bf 110: 12Lv; 6Lo; VC (e)	Two Fw 190: 12H; 1:30Lv Two Bf 109: 6Lv; 6Lo
56	Bf 109: 12H It: MC.205: 3H Hu: Bf 109: 3H Ro: IAR 81: 3H (r)	Three Fw 190: 3Lo; 12H; 12Lv	Four Bf 109: 12H; 12Lv; 12Lo; 10:30Lv
61	Bf 109: 9H It/Hu/Ro: Bf 109: 9Lo	Two Bf 109: 9Lv; 12H It: MC.205: 3H Hu/Ro: Bf 109: 3H	Three Bf 109: 12Lv; 12H; 10:30Lv Bf 110: 12Lo
62	Two Fw 190: 12H; 1:30H It: Bf 109: 10:30H Hu: Bf 109: 3H Ro: IAR 81: 6Lv (r)	Two Fw 190: 12H; 1:30H It: Bf 109: 10:30Lv Hu: Heja II: 6H Ro: IAR 81: 6H (r)	Four Fw 190: 12Lv; 1:30Lv; 3Lo; 9H
63	Three Fw 190: 12H; 12Lv; 3Lo	Three Fw 190: 6Lv; 1:30Lv; 12H	Five Fw 190: 12H; 10:30Lv; 9Lv; 6H; VD (b)
64	Three Bf 109: 1:30H; 12Lo; VD (d)	Four Fw 190: 12Lo; 3Lv; 1:30H; 9H	Five Fw 190: 12H; 1:30H; 3H; 6H; 9H
65	Three Bf 109: 10:30H; 12H; 3H	Four Fw 190: 6H; 12H; 12Lv; 6Lo	Five Bf 109: 12Lv; 3H; 1:30H; 6H; VD (b)
66	Random Event		

- b Bomber cannot fire on this fighter, nor can the escort remove it.
- c Fighters are driven off by other bombers. Re-roll if out of formation.
- d Top Turret and Radio Room MG may fire. Escort cannot remove this attacker.
- e Only the Ball Turret may fire. Escort can remove this attacker.
- f German bombs dropped on the formation from above. Roll 1d6:
1-5 = 1 Light Flak hit 6 = 1 Heavy flak hit All results create Disrupted Formation
- g German Rockets lobbed into the formation from aft. Roll 1d6:
1-3 = Miss 4-5 = 1 light Flak hit 6 = 1 Heavy Flak hit All results create Disrupted Formation
- h If over Germany, substitute Ju 88C.
- r Note: all IAR-80 and 81s were withdrawn from combat against the USAAF after 20th July 44 due to high losses. After this date, treat as Bf 109.

AXIS AIRCRAFT CODES

Table 5-10* - AXIS AIRCRAFT TYPE - REGENSBURG						
Aircraft	D10	Code		Aircraft	D10	Code
Bf 109	1-3	A		G.50	All	I-1
	4-9	B		G.55	All	I-4
	10	C		MC.200	1-8	I-1
Bf 110	1-4	D			9-10	I-2
	5-10	G		MC.202	All	I-2
Fw 190	1-7	J		MC.205	1-8	I-2
	8-10	K			9-10	I-3
Ju 88	All	H		Re.2001	All	I-2
Me 210	All	E		Re 2002	All	I-2
Me 410	1-9	E		Re.2005	All	I-5
	10	K		Bf 109 (It)	All	A

Table 5-10** - AXIS AIRCRAFT TYPE - SUMMER 44 - SOUTHERN EUROPE						
Aircraft	D10	Code		Aircraft	D10	Code
GERMAN				ITALIAN		
Bf 109	1-9	B		G.55	All	I-4
	10	D		MC.205	1-7	I-2
Bf 110	1-9	G			8-10	I-3
	10	L		Re.2005	All	I-5
Me 410	1-6	E		Bf 109 (It)	1-2	A
	7-9	G			3-9	B
	10	K			10	U-7
Fw 190	1	J				
	2-10	L				
Ju 88	All	H				
HUNGARIAN				ROMANIAN		
Heja II	All	I-1		IAR 81	All	R-4
Bf 109 (Hu)	1-3	A		Bf 109 (Ro)	1-3	A
	4-10	B			4-10	B
Me 210	All	E				

TABLE 5-11 HITS BY WEAPON CODE - ITALIAN

I-1 2 x 12.7mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1		1		1	
1	0		0		1	
2	0		1		1	
3	1		1		1	
4	1		1		1	
5	1		1		1	
6	1		1		2	
7	1		1		2	
8	1		1		2	
9	1		2		2	
≥ 10	1		2		2	

I-2 2 x 12.7mm and 2 x 7.7mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1		1		1	
1	0		1		1	
2	0		1		2	
3	1		1		2	
4	1		1		2	
5	1		1		2	
6	1		2		2	
7	2		2		3	
8	2		2		3	
9	2		2		3	
≥ 10	2		3		3	

I-3 2 x 20mm and 2 x 12.7mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1	3	1	3	1	3
1	0	0	1	0	1	1
2	0	0	1	0	1	1
3	1	0	0	1	1	1
4	0	1	0	1	1	1
5	1	1	1	1	2	1
6	1	1	1	1	2	1
7	1	1	1	1	2	1
8	1	1	1	1	1	2
9	1	1	2	1	1	2
≥ 10	1	1	2	2	2	2

I-4 3 x 20mm and 2 x 12.7mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1	3	1	3	1	3
1	0	0	0	1	1	1
2	0	1	1	1	1	1
3	1	1	1	1	2	1
4	1	1	1	1	1	2
5	1	1	1	1	1	2
6	1	1	1	1	1	2
7	1	1	2	1	2	2
8	1	1	1	2	2	2
9	1	1	1	2	2	3
≥ 10	1	2	2	2	2	3

I-5 5 x 20mm and 2 x 12.7mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1	3	1	3	1	3
1	0	1	0	1	1	2
2	1	1	1	1	1	2
3	1	1	0	2	2	2
4	1	1	1	2	1	3
5	0	2	1	2	2	3
6	1	2	1	2	2	3
7	1	2	1	2	1	4
8	1	2	1	3	2	4
9	1	2	2	3	2	4
≥ 10	2	2	2	4	3	4

Modifiers: +1 VD +1 Experte
 +1 VC -1 Novice

TABLE 5-11 HITS BY WEAPON CODE - ROMANIAN/HUNGARIAN

R-1 6 x 7.9mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1		1		1	
1	0		1		1	
2	1		1		2	
3	1		1		2	
4	1		1		2	
5	1		2		2	
6	1		2		2	
7	1		2		3	
8	2		2		3	
9	2		2		3	
≥ 10	2		2		3	

R-2 2 x 13mm and 4 x 7.9mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1		1		1	
1	0		1		2	
2	1		1		2	
3	1		1		2	
4	2		2		3	
5	2		2		3	
6	2		2		3	
7	2		2		3	
8	2		3		4	
9	2		3		4	
≥ 10	2		4		5	

R-3 2 x 20mm FF and 2 x 7.9mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1	3	1	3	1	3
1	0	0	0	0	1	0
2	0	0	1	0	0	1
3	1	0	1	0	0	1
4	1	0	0	1	1	1
5	1	0	0	1	1	1
6	0	1	0	1	1	1
7	0	1	1	1	1	1
8	0	1	1	1	1	1
9	1	1	1	1	1	1
≥ 10	1	1	1	1	1	2

R-4 2 x 20mm and 4 x 7.9mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1	3	1	3	1	3
1	0	0	1	0	1	1
2	0	0	1	0	1	1
3	1	0	0	1	1	1
4	0	1	0	1	1	1
5	1	1	1	1	2	1
6	1	1	1	1	2	1
7	1	1	1	1	2	1
8	1	1	1	1	1	2
9	1	1	2	1	1	2
≥ 10	1	1	2	2	2	2

H-1 2 x 20mm FF and 2 x 8mm						
DR	12 / VD 10:30/1:30		3 / 9 / VC		6	
	1	3	1	3	1	3
1	0	0	0	0	1	0
2	0	0	1	0	0	1
3	1	0	1	0	0	1
4	1	0	0	1	1	1
5	1	0	0	1	1	1
6	0	1	0	1	1	1
7	0	1	1	1	1	1
8	0	1	1	1	1	1
9	1	1	1	1	1	1
≥ 10	1	1	1	1	1	2

Modifiers: +1 VD +1 Experte
 +1 VC -1 Novice

APPENDIX 1 - FIREPOWER CODES

These are aircraft which are included in the supplementary aircraft codes. Several of these (MC.200, He 112B) did not see action in the timescale of these missions, but I have included them so you can use them in earlier periods.

Code	Weapons Fit	Aircraft	
I-1	2 x 12.7mm	G.50bis MC.200 (2 gun)	Re.2000 Heja II
I-2	2 x 12.7mm 2 x 7.7mm	MC.200 (4 gun) MC.202 MC.205 Ser I	Re.2001 Re.2002
I-3	2 x 20mm 151/20 2 x 12.7mm	MC.205 Ser III	
I-4	3 x 20mm 151/20 2 x 12.7mm	G.55 Ser I	
I-5	5 x 20mm 151/20 2 x 12.7mm	Re.2005	
R-1	6 x 7.9mm	IAR 80A	
R-2	2 x 13mm 4 x 7.9mm	IAR 80B	
R-3	2 x 20mm FF/M 2 x 7.9mm	IAR 80C He 112B-1	
R-4	2 x 20mm 151/20 4 x 7.9mm	IAR 81C	
H-1	2 x 20mm FF/M 2 x 7.9mm	He 112B-1	

Use these tables if you aren't using the new Axis "aircraft codes" for their attacks.

5-11 HIT EFFECT MULTIPLIER								
d6	He 112	IAR 80						
	1-2	1-6						
1	0.5 (frd)	0.5 (frd)						
2	0.5 (fru)	0.5 (fru)						
3	0.5 (fru)	0.5 (fru)						
4	1	1						
5	1	1						
6	1	1						

5-11 HIT EFFECT MULTIPLIER								
d6	G.50	G.55	MC.200	MC.202	MC.205	Re.2000	Re.2001	Re.2005
	1-2	4-6	1-2	1-2	2-6	1-2	1-2	2-6
1	0.5 (frd)	0.5 (fru)	0.5 (frd)	0.5 (frd)	0.5 (frd)	0.5 (frd)	0.5 (frd)	0.5 (frd)
2	0.5 (frd)	0.5 (fru)	0.5 (frd)	0.5 (fru)	0.5 (fru)	0.5 (frd)	0.5 (fru)	0.5 (fru)
3	0.5 (fru)	1	0.5 (fru)	0.5 (fru)	0.5 (fru)	0.5 (fru)	0.5 (fru)	1
4	0.5 (fru)	1	0.5 (fru)	1	1	0.5 (fru)	1	1
5	1	1	1	1	1	1	1	2
6	1	2	1	2	2	1	1	2

Treat a '0' result as Superficial Damage.

frd = fractions rounded down

fru = fractions rounded up